

**Business Analytics, Applied Modelling
and Prediction (ST 2187)**

Coursework project 2025



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Executive Summary

This report contains several insights obtained from data visualisation in Tableau. These insights provide a clear view on certain products that are underperforming and the circumstances around them. The insights that I have gathered are as follows:

- a) Tables are overwhelmingly more profitable in certain countries than others.
- b) Tables are the least profitable product in the current product lineup.
- c) Tables have the highest discount applied to them as compared to other products.
- d) Although sales of tables have been increasing, the profitability of tables has been constantly decreasing.
- e) Tables have higher volume of sales in certain seasons every year.

This emphasis of this report is on Tables since it is the product with the least profitability in the product lineup.

These insights may help the company by making tables a profitable product in their product lineup. Tables are seen to have the highest discount among other products. Therefore, this could lead to the high discount affecting the profitability of tables.

Some countries are more profitable for tables than others. This can potentially be balanced by adjusting the discount the countries get. For instance, the countries with high profitability on tables may enjoy a slightly lesser discount and the countries with low profitability can enjoy higher discount to incentivise purchases and to balance the overall profitability.

It is clear that certain seasons see an uptick in sales of tables. More promotions or special discounts can be in place in these time periods to encourage people to buy tables.

In this report, the visualisations created can help to increase the company's profit margins and to compete with competitors.

Introduction

This report is based on data collected by the company from January 2020 to January 2024. It includes the sales and profit figures for the product lineup sold by the company. It includes data on which regions they have been shipped to, the discount of the products, and customer particulars, among others.

This report aims to use the data visualisation from Tableau to provide insights on product performance and to improve the company profits.

Story Insights

3a) Profit and sales overview of products

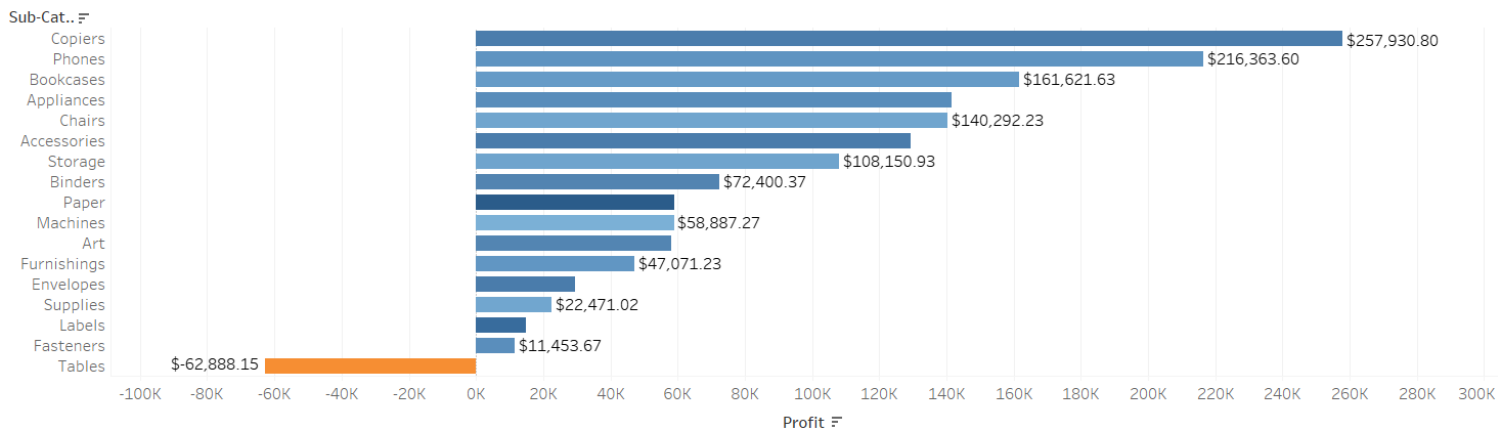


Figure 1: The profitability of the products

Tables are the worst performing product in the company's lineup in profit. They are the least profitable in the product lineup and have incurred a loss of \$62,888.15 in the time period.

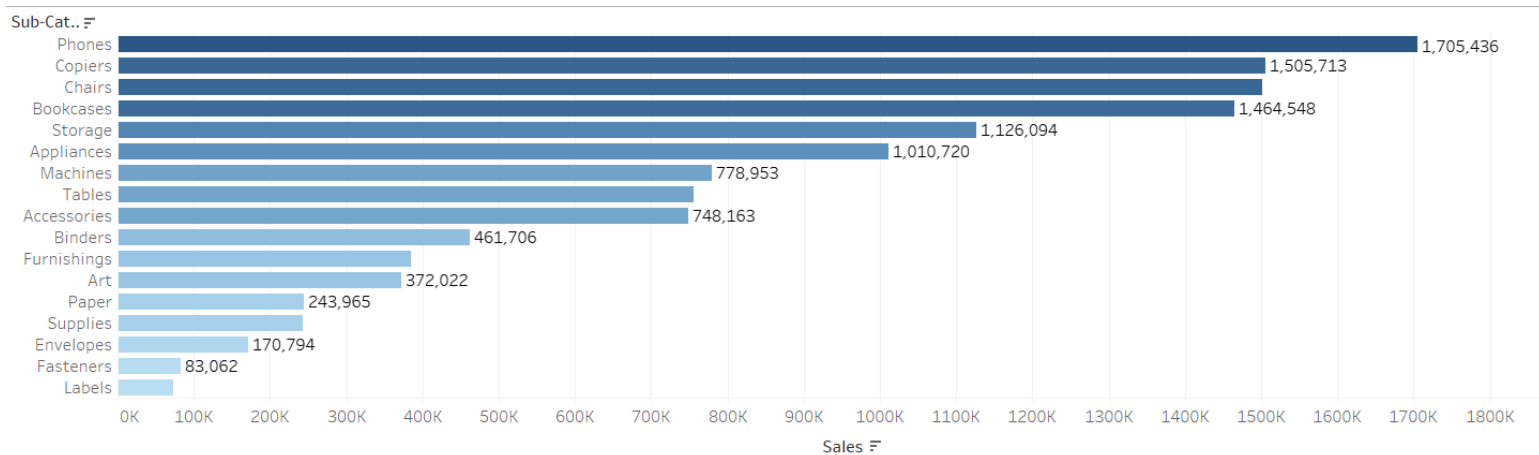
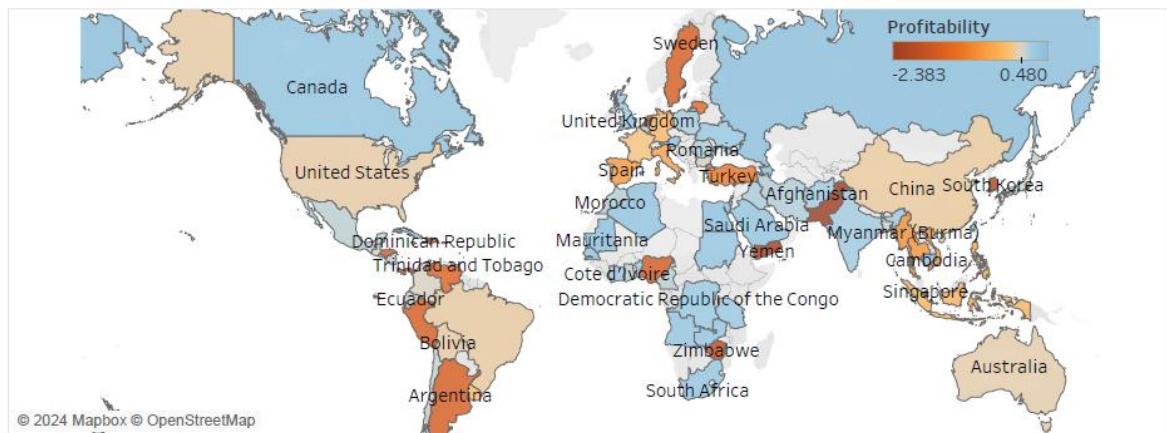


Figure 2: The total sales of the products

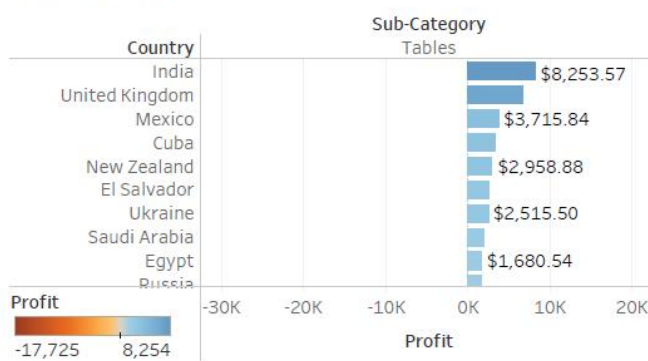
Tables are the 8th highest selling product and have made \$755,993 in sales. This shows that the lack of profitability of tables is not due to a lack of sales. The reasons will be explored further in this report with other insights.

3b) Profitability of tables across regions

Profitability of Tables across countries



Most Profitable Countries For Tables



Least Profitable Countries For Tables

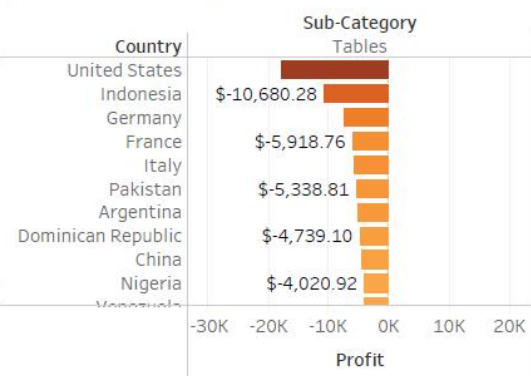


Figure 3: This shows how the profitability of tables across regions

Although we have established that tables are the least profitable product in the company lineup, this data visualisation shows that tables are not making losses across all regions. For example, in countries like India and the United Kingdom, tables are making a profit of \$8253.57 (India). However, in the same visualisation, we can see that tables are not profitable in countries such as the United States and Indonesia making losses such as \$10,680.28 in the case of Indonesia.

3c) Tables have the highest discount in products

Discount for Table

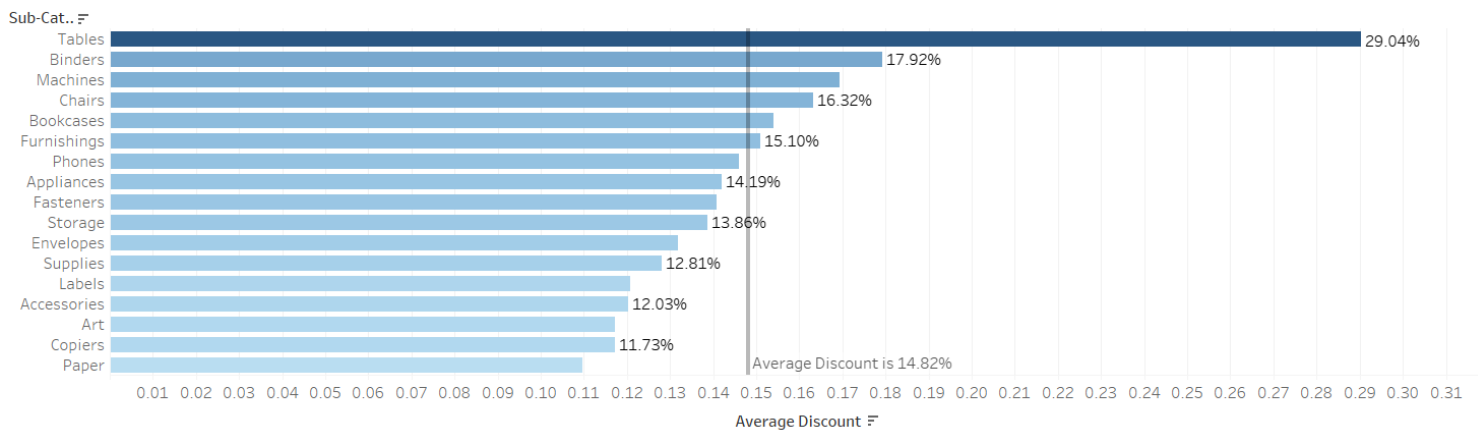


Figure 4: The discount of tables vs other products

In figure 4, we can clearly see that the discount of tables far exceeds the average discount of other products. The tables discount sits at 29.04% while the average discount is at 14.82%. There is a **difference of 14.22%**. Tables have almost double the average discount as compared to the average discount offered by the company for other products. This high discount contributes to the poor profitability of tables. This is why tables incur losses even though it is selling very well as compared to other products. The next 2 most discounted products, binders and machines are making a tidy profit for the company despite being discounted above average as well.

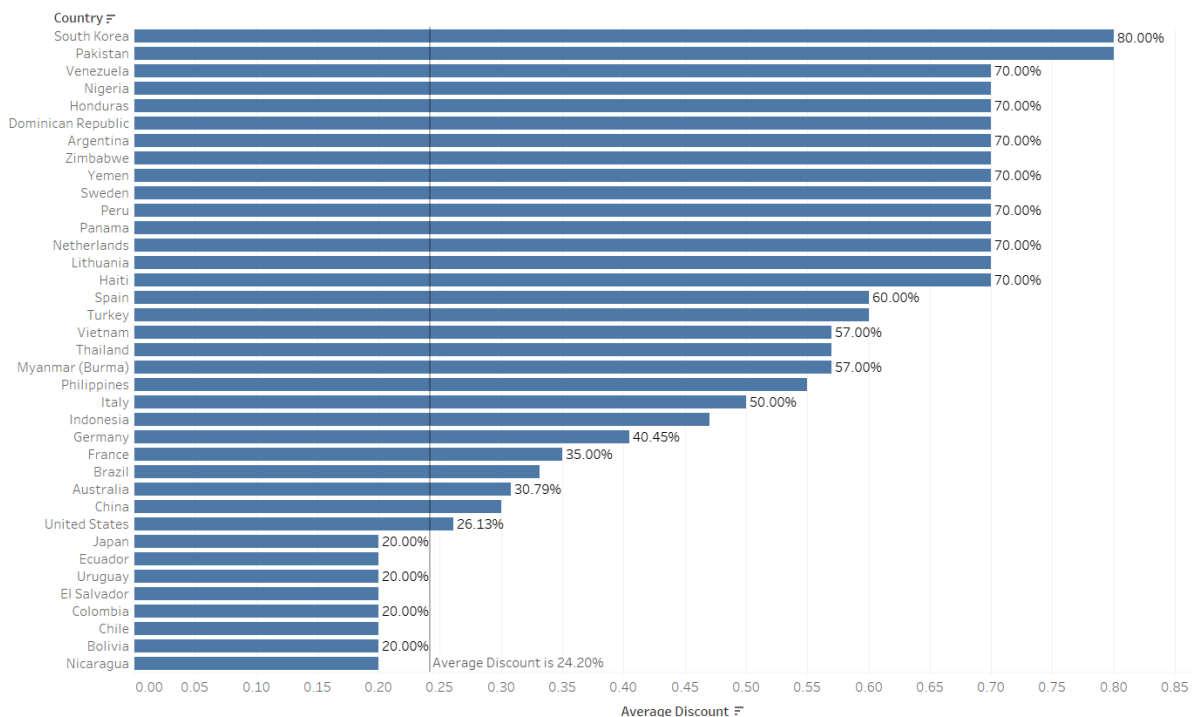


Figure 5: Average discount of tables across countries

Figure 5 shows that countries like South Korea and Pakistan enjoy the highest discounts for tables at 80%, closely followed by Venezuela and Nigeria who have a 70% discount. In fact, there are quite a number of countries that have a 70% discount on tables. However, it has been calculated that the average discount across all countries is only at 24.20%. This means that a sizeable portion of the countries that buy tables are having almost triple the discount as other countries.

This can explain why some countries are more profitable for tables as compared to others.

3d) Table performance by seasonality

Profit VS Sales of Tables



Figure 6: Profit VS Sales of tables

Figure 6 shows that despite the sales of tables increasing year after year since 2020 to 2023, the losses made on tables grows more and more as well. This can be attributed to the reasons as discussed above. If the company makes adjustments similarly to some suggestions given towards the end, the strong sales of the tables may help the company to make a good profit since tables are one of the top selling products in the lineup currently.

Seasonality of Profit & Sales

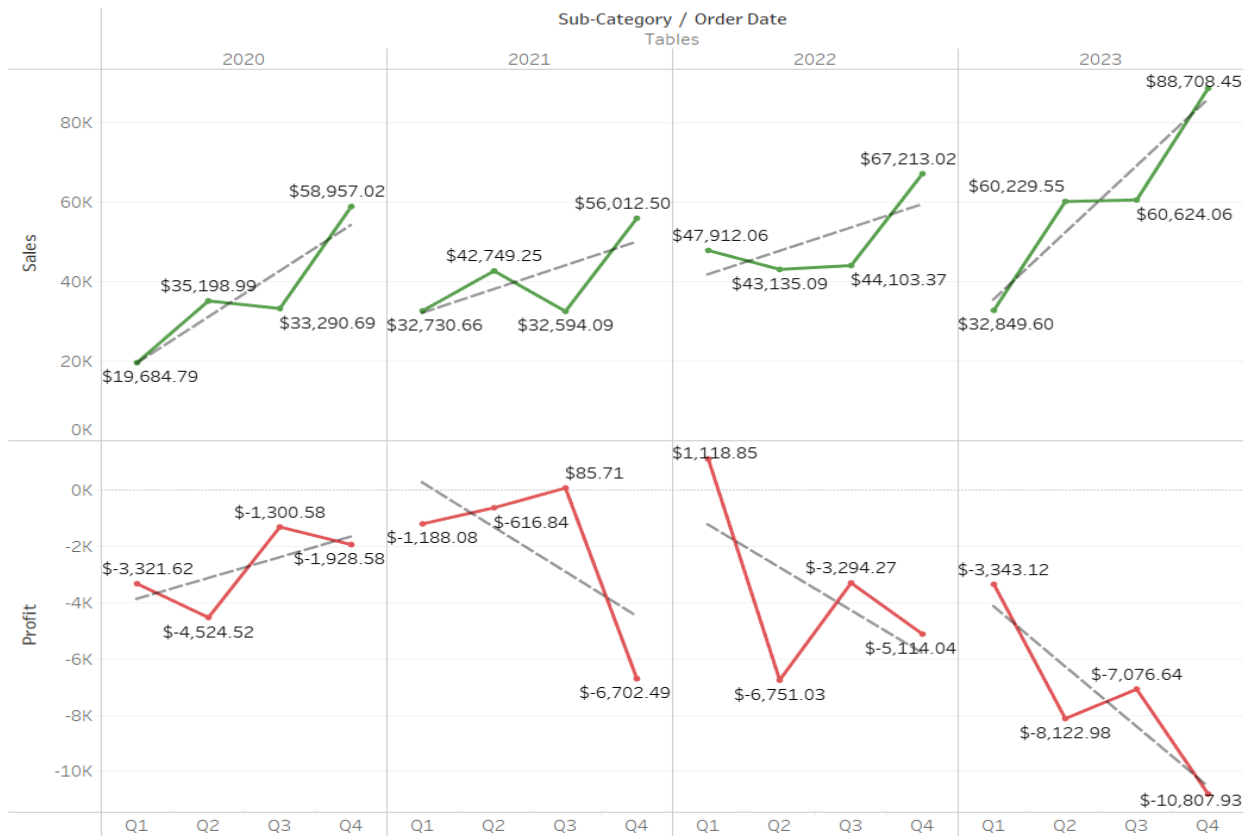


Figure 7: Seasonality of tables performance

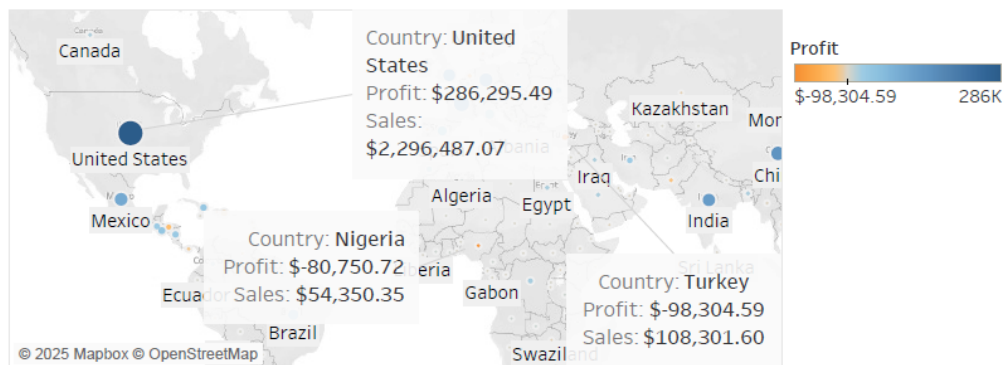
Figure 7 shows the sales and profit comparison of tables between the years 2020 to 2023. It shows that the sales of tables has been improving every year. For example, in 2020, the sales of tables amounted to \$58,957.02 in the fourth quarter of 2020. In 2023, it had risen to \$88,708.45 for the same fourth quarter period. However at the same time, the losses have been getting higher for tables despite strong sales. In 2020, a loss of \$1,928.58 was recorded in the fourth quarter of 2020 while in 2023, it the loss had increased to \$10,807.93 in the same fourth quarter. However, it is important to note that there is always a constant uptick in the same quarter from 2020 to 2023. The second and fourth quarter of every year usually has a surge in tables purchased. This shows that the pattern can be used to increase the profits with some correct steps during those 2 quarters and can be used to study why the first and third quarter has a decline in interest.

Another observation to note is that the greater the sales, the greater the loss. For instance, in Q4 of 2020, the sales hit \$58,957.02 while the loss was \$1,928.58. When compared to Q4 of 2023, the sales hit \$88,708.45 while the loss became \$10,807.93. This shows that the greater the number of sales, the

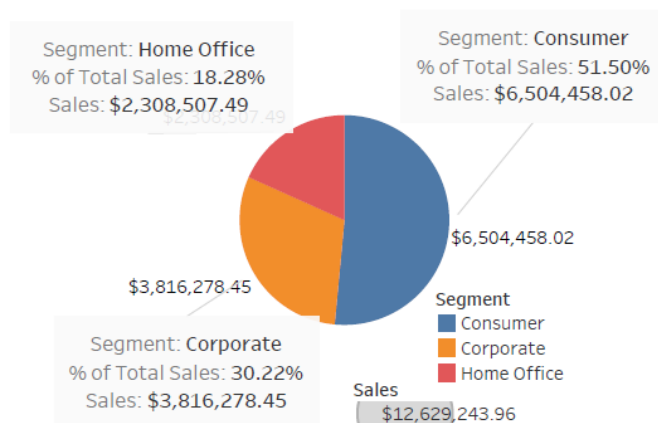
greater the loss experienced by the company. This can be attributed to the discounts that we explored earlier. The heavy discounts offset the good sales and it becomes very difficult to turn a profit.

3e) Overview of sales and profit across countries

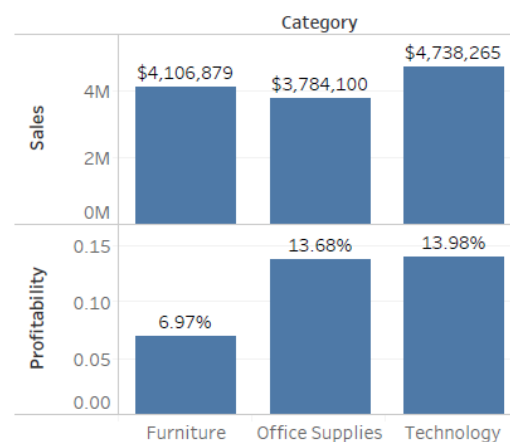
Country vs Profit



Sales By Segment



Sales & Profitability by Category



When comparing the profit from all countries, the United States is the leader with \$286,295.49 in profit generated while Turkey was the country that incurred the heaviest losses at \$98,304.59. When analysing by segment, consumers had the lion's share of the total number of sales generated with 51.5% , followed by corporate and home office which had 30.22% and 18.28% respectively.

When analysing the category section, technology had the highest sales and it the most profitable. However, we must note that furniture, despite being the second highest in sales numbers at \$4,106,879, is only at 6.97% profitability. This is not considered healthy since profit margins above 10% are considered to be good and progressive for the company.

Conclusion

In conclusion, the report has highlighted that tables are the least profitable product in the product lineup. This is the outcome despite having strong sales. Tables are also not as profitable in some countries as compared to others. The losses incurred by the table is due to the heavy discount in place for it. It is the most discounted product in the product lineup by quite an amount and it can lead to the company struggling to turn a profit on tables despite having strong sales numbers. The USA has generated the highest profit for the company while Turkey has the highest losses incurred. There are some recommendations to help alleviate the problem with the tables.

Recommendations

5a) Adjusting of discounts

Since the discount of the table is what's causing the profit of the tables to be affected, we should look into addressing that. As seen in Figure 4 above (and attached below), the discount for tables is too high to prevent losses, let alone turning a good profit. However, to drastically slash the discount of the tables will cause the sales of tables to decline, resulting in even further losses for the company.

Discount for Table

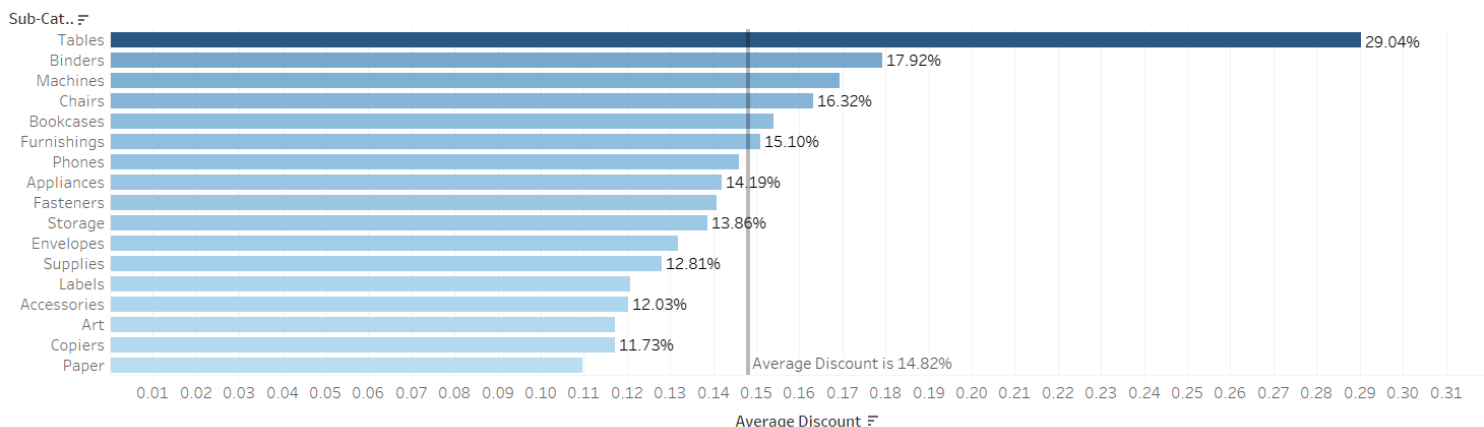
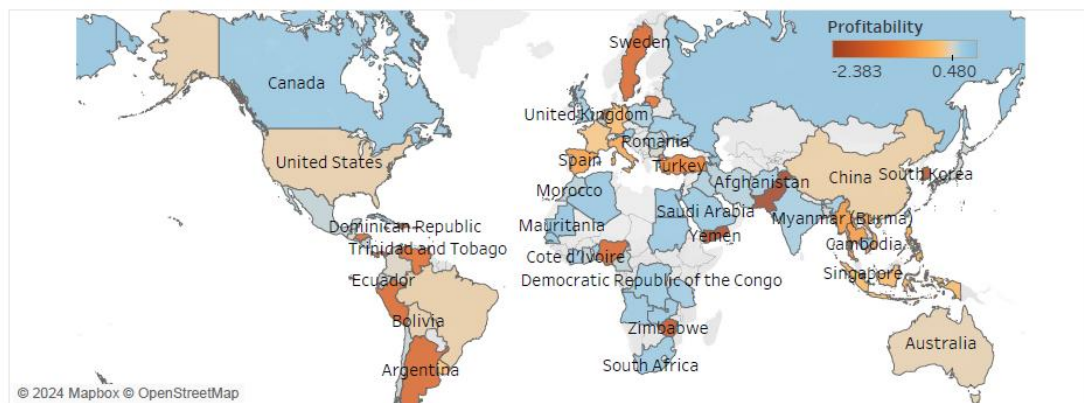


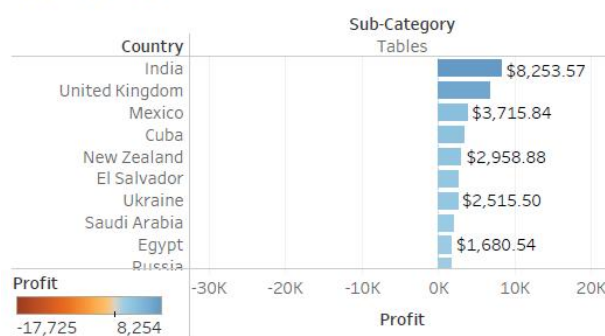
Figure 4

In order to circumvent this, the company may choose to adjust the amount of discount it gives to certain countries. For instance, in figure 3 (attached below as well), we can see that some countries are more profitable than others when it comes to tables. The company may adjust the discounts to be lesser in already profitable countries and can increase the discounts in the less profitable countries in order to encourage other markets to buy more tables. This could lead to them incurring lesser losses in certain countries such as the USA and Indonesia while still getting some sales in the countries they are already profitable at.

Profitability of Tables across countries



Most Profitable Countries For Tables



Least Profitable Countries For Tables

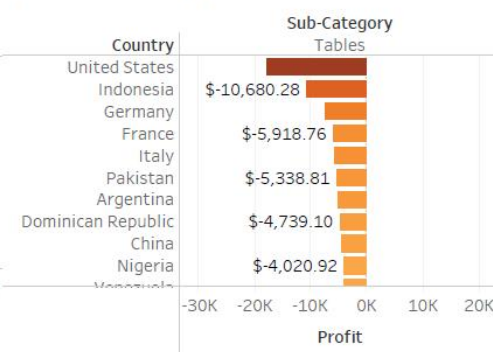


Figure 3

5b) More promotions in the year

As seen in figure 7 (attached below as well), there are certain periods of the year where the sales of tables increase a lot. If the company capitalises on these periods, there will be lower losses and potentially may even turn a profit. For instance, in Q2 and Q4 of every year, the data shows that the sales rises by a noticeable amount while it decreases in Q1 and Q3. This could be due to schools buying tables during the school term breaks or end of the year breaks to replace older tables. They can sell chairs and tables as a set to incentivise those who prefer more value for their money. They can offer bulk purchase discounts to the schools if they buy in a large supply for their classrooms. This will encourage schools to choose this company as the priority supplier for furniture and other school supplies that the company sells. This can in turn increase the profits that the company may obtain through their products.

The company may also prioritise heavier discounts in these periods to incentivise purchases and to offset the financial loss, they can reduce the discounts in other months of the year.

Seasonality of Profit & Sales

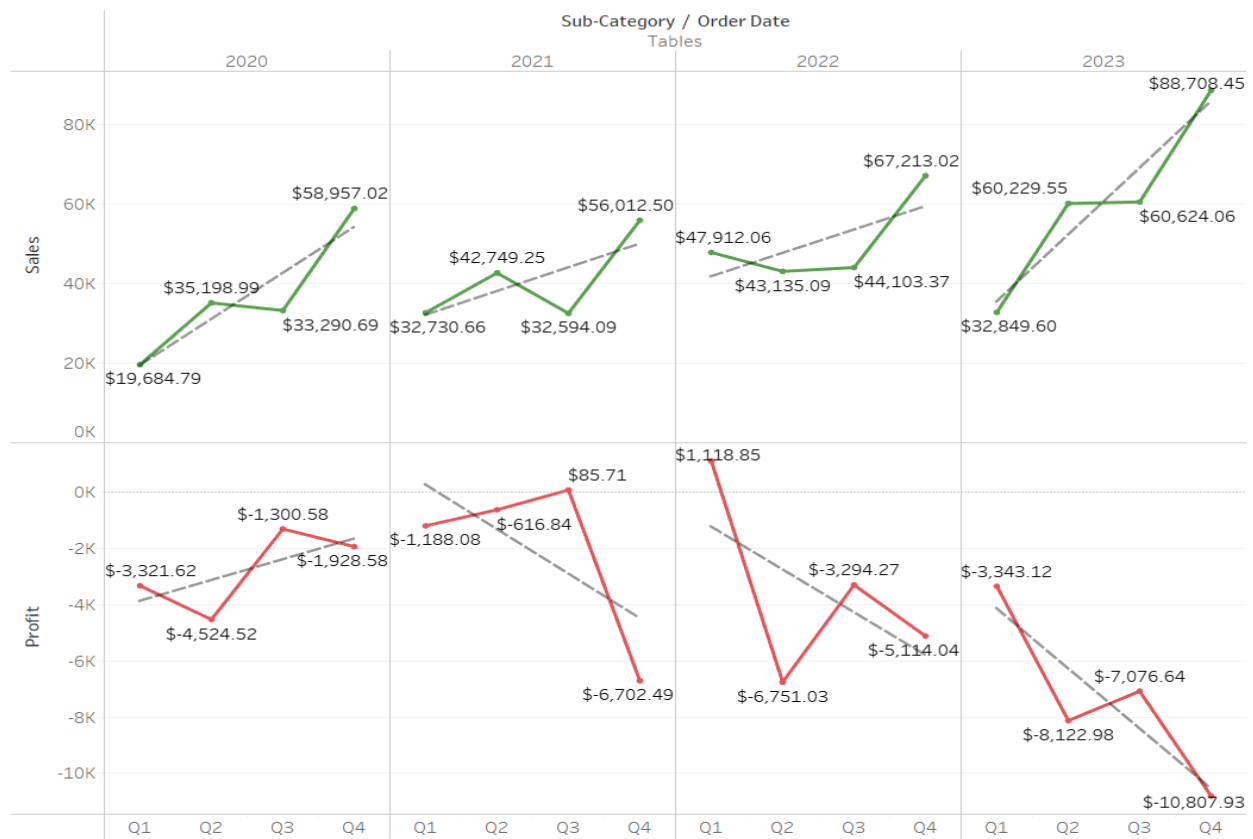


Figure 7

5c) Loyalty programs

The company can look into introducing loyalty programs for repeat customers. This can be in the form of further discounts for repeat customers which regular customers would not have. This can encourage clients to keep supporting the company on their products since they will receive an incentive for the continued patronage.

References

What is a good profit margin?

~ <https://www.brex.com/journal/what-is-a-good-profit-margin>